PROBLEM STATEMENT NO. 22

Interview Trainer Agent

AI-powered, personalized interview preparation using IBM Granite and RAG

1. Problem Statement

Preparing for job interviews is a major challenge for students, fresh graduates, and experienced professionals. Many candidates struggle to identify relevant interview questions, prepare effective answers, explain their technical knowledge, and confidently handle HR and behavioral questions.

Existing interview preparation methods often provide generic questions that do not match an individual's educational background, experience level, skills, or targeted job role.

The **Interview Trainer Agent** aims to solve this problem using Artificial Intelligence and Retrieval-Augmented Generation (RAG). Users can upload their resume or enter their profile details, skills, experience level, and desired job role. Based on this information, the agent retrieves relevant interview questions, technical concepts, industry expectations, and HR scenarios from a knowledge base.

The system generates personalized interview questions, sample answers, coding challenges, behavioral questions, and practical preparation strategies. It also evaluates user responses and provides constructive feedback, highlighting strengths, weaknesses, and areas for improvement.

The solution supports both technical and non-technical interview preparation, helping candidates improve their communication skills, problem-solving abilities, technical knowledge, and confidence.

By providing personalized, interactive, and AI-powered interview preparation, the Interview Trainer Agent helps candidates become better prepared for competitive job opportunities.

2. Main Objectives

Objective	Description
Resume Analysis	Analyze uploaded resumes and identify skills, qualifications, and experience.
Personalized Questions	Generate interview questions based on the selected job role and experience level.
Technical Preparation	Provide programming, database, aptitude, and domain-specific questions.
HR Preparation	Generate HR and behavioral interview questions.
AI-Generated Answers	Provide sample answers with clear explanations.
Answer Evaluation	Analyze user responses and provide scores and improvement suggestions.
RAG Integration	Retrieve relevant information from a knowledge base to generate accurate and context-aware responses.
Mock Interviews	Simulate real interview sessions with interactive questions and feedback.

3. Key Features

Feature	Description
Resume Analyzer	Extracts skills and qualifications from resumes.
Job Role Selection	Supports different job positions.
Technical Interview	Generates role-specific technical questions.
HR Interview	Provides behavioral and HR questions.
Mock Interview	Simulates an actual interview.
Answer Evaluation	Evaluates answers and provides feedback.
Model Answers	Generates sample answers.
Performance Report	Shows strengths and improvement areas.

4. Technology Stack

Technology	Purpose
IBM Granite	Mandatory foundation model for AI-generated interview questions, answers, and feedback.
IBM Cloud Lite	Cloud-based deployment and IBM services.
Python	Primary programming language.
LangChain	LLM orchestration and RAG pipeline integration.
ChromaDB or FAISS	Vector storage and semantic retrieval.
Flask or FastAPI	Backend API and application logic.
Streamlit or React	User interface.
PyPDF2 or PyMuPDF	Resume and document text extraction.
SQLite or PostgreSQL	User profiles and application data storage.

5. Proposed System Workflow

1. User opens the application.
2. User enters profile details or uploads a resume.
3. The system analyzes the resume and profile.
4. User selects a job role and experience level.
5. RAG retrieves relevant information from the knowledge base.
6. IBM Granite generates personalized interview content.
7. The agent presents questions and the user submits answers.
8. AI evaluates the responses.
9. The system provides feedback and improvement suggestions.

10. A final interview performance report is generated.

6. Example Use Case

Sample User Profile

Name	Kartik Malve
Education	Bachelor of Computer Science
Experience	Fresher
Target Role	Python Developer
Skills	Python, Java, C, MySQL, PostgreSQL

Expected Agent Output

• 10 Python technical interview questions.

• 5 SQL and database questions.

• 5 HR and behavioral questions.

• Python coding challenges.

• Sample answers with explanations.

• Interactive mock interview.

• Personalized performance evaluation.

7. Expected Outcome

The final system will be an AI-powered Interview Trainer Agent that provides personalized interview preparation based on a candidate's profile, resume, experience, and targeted job role. It will use IBM Granite and RAG to retrieve relevant knowledge, generate meaningful questions, evaluate answers, and provide actionable feedback.

The solution will make interview preparation more accessible, interactive, and effective for students, freshers, and working professionals.